15.4: Alterations of chromosome number or structure cause some genetic disorders

As you have learned so far in this chapter, the phenotype of an organism can be affected by small-scale changes involving individual genes. Random mutations are the source of all new alleles, which can lead to new phenotypic traits.

Large-scale chromosomal changes can also affect an organism's phenotype. Physical and chemical disturbances, as well as errors during meiosis, can damage chromosomes in major ways or alter their number in a cell. Large-scale chromosomal alterations in humans and other mammals often lead to miscarriage of a fetus, and individuals born with these types of genetic defects commonly exhibit various developmental disorders. Plants appear to tolerate such genetic defects better than animals do.

Interview

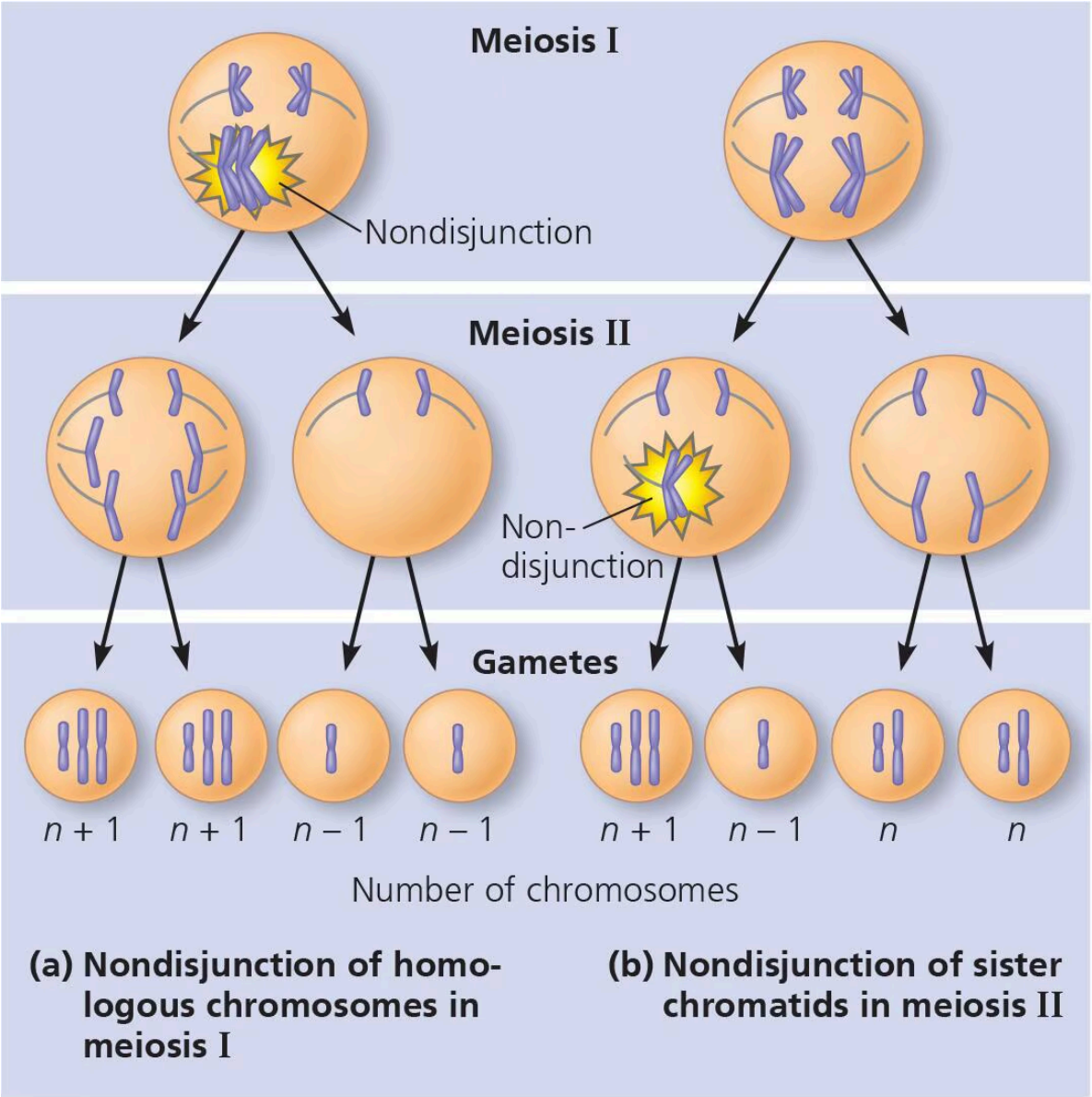
Interview with Terry Orr-Weaver: Why are humans so bad at meiosis?

Abnormal Chromosome Number

Ideally, the meiotic spindle distributes chromosomes to daughter cells without error. But there is an occasional mishap, called a **nondisjunction**, in which the members of a pair of homologous chromosomes do not move apart properly during meiosis I or sister chromatids fail to separate during meiosis II **(Figure 15.13)**. In nondisjunction, one gamete receives two of the same type of chromosome and another gamete receives no copy. The other chromosomes are usually distributed normally.

Figure 15.13

Meiotic nondisjunction.



Part a. Nondisjunction of homologous chromosomes in meiosis I. One homologous pair fails to separate; resulting gametes contain too many or two few chromosomes: n plus1, n plus1, n-1 and n-1.

- Parent cell shows two pairs of homologous chromosomes in anaphase I. The top pair are separated normally by the spindle fibers, but the bottom pair have stuck together in a nondisjunction, and are moving together towards the same pole of the cell rather than to opposite poles as in normal anaphase I.
- During meiosis II, one daughter cell has only one set of sister chromatids splitting during anaphase II, and the other has three sets of sister chromatids splitting during anaphase II.
- The resulting gametes all have too many or two few chromosomes. Half the gametes contain three chromosomes and are n plus1, and half contain one chromosome and are n-1.

Part b. Nondisjunction of sister chromatids in meiosis II. Meiosis I proceeds normally, but sister chromatids fail to separate during meiosis II and the resulting gametes are n plus1, n-1, n and n.

- Parent cell shows two pairs of homologous chromosomes separating normally during anaphase I of meiosis I.
- During anaphase II of meiosis II, one set of sister chromatids in one daughter cell fails to separate, and instead moves together to the same pole of the cell in a nondisjunction. The other set of sister chromatids in that cell, and both sets of sister chromatids in the other daughter cell all separate normally.
- The resulting gametes have chromosome numbers as follows: 3 (n plus1), 1 (n-1), 2 (n) and 2 (n).

Gametes with an abnormal chromosome number can arise by nondisjunction in either meiosis I or meiosis II. For simplicity, the figure does not show the spores formed by meiosis in plants. Ultimately, spores form gametes that have the defects shown. (See Figure 13.6.)

If either of the aberrant gametes unites with a normal one at fertilization, the zygote will also have an abnormal number of a particular chromosome, a condition known as **aneuploidy**. Fertilization involving a gamete that has no copy of a particular chromosome will lead to a missing chromosome in the zygote (so that the cell has 2n−1 chromosomes); the aneuploid zygote is said to be **monosomic** for that chromosome. If a chromosome is present in triplicate in the zygote (so that the cell has 2n+1 chromosomes), the aneuploid cell is **trisomic** for that chromosome. Mitosis will subsequently transmit the anomaly to all embryonic cells. Monosomy and trisomy are estimated to occur in 10–25% of human conceptions and are the main reason for pregnancy loss. If the organism survives, it usually has a set of traits caused by the abnormal dose of the genes associated with the extra or missing chromosome. Down syndrome is an example of trisomy in humans that will be discussed later. Nondisjunction can also occur during mitosis. If such an error takes place early in embryonic development, then the aneuploid condition is passed along by mitosis to a large number of cells and is likely to have a substantial effect on the organism.

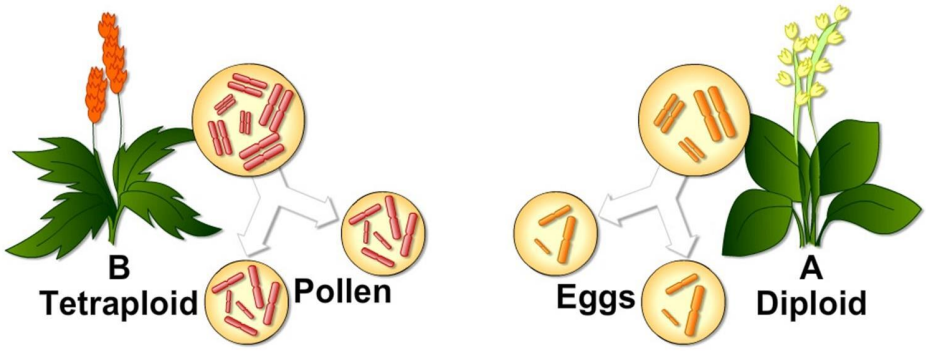
Some organisms have more than two complete chromosome sets in all somatic cells. The general term for this chromosomal alteration is **polyploidy**; the specific terms *triploidy* (3n) and *tetraploidy* (4n) indicate three and four chromosomal sets, respectively. One way a triploid cell may arise is by the fertilization of an abnormal diploid egg produced by nondisjunction of all its chromosomes. Tetraploidy could result from the failure of a 2n zygote to divide after replicating its chromosomes. Subsequent normal mitotic divisions would then produce a 4n embryo.

Polyploidy is fairly common in the plant kingdom. The spontaneous origin of polyploid individuals plays an important role in plant evolution (see Concept 24.2). Many species we eat are polyploid: Bananas are triploid, wheat hexaploid (6n), and strawberries octoploid (8n). Polyploid animal species are much less

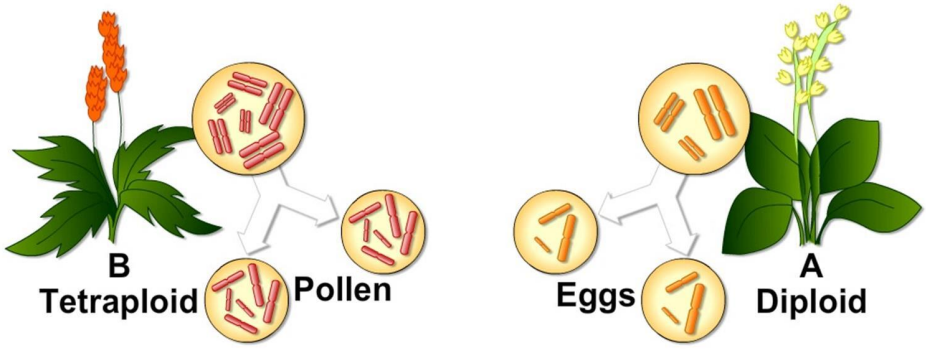
common, but there are a few fishes and amphibians known to be polyploid. In general, polyploids are more nearly normal in appearance than aneuploids. One extra (or missing) chromosome apparently disrupts genetic balance more than does an entire extra set of chromosomes.

Animation

Polyploid Plants



Play



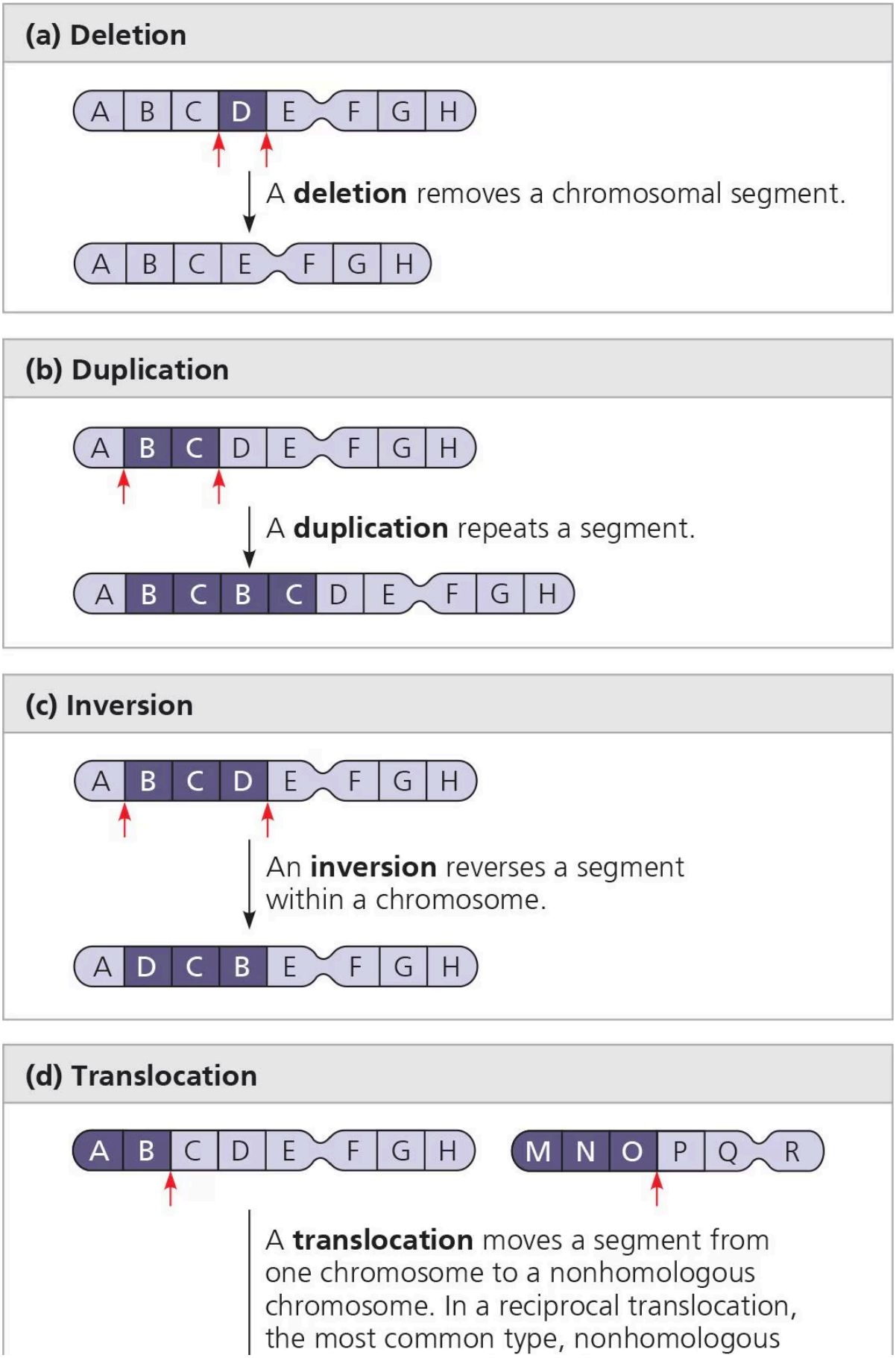
Play

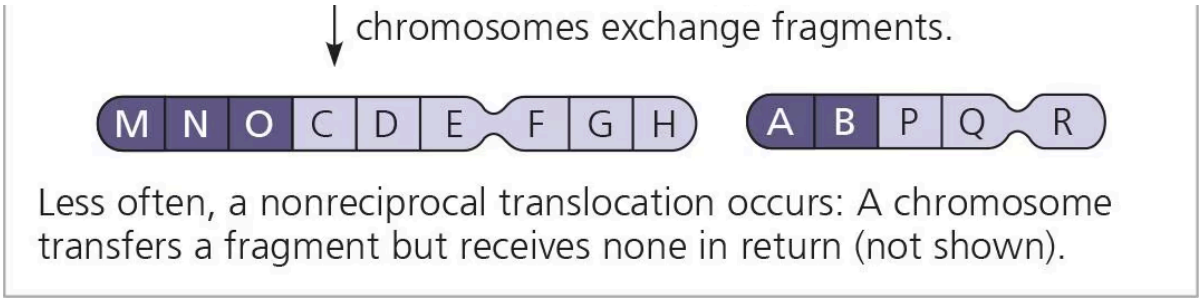
Alterations of Chromosome Structure

Errors in meiosis or damaging agents such as radiation can cause breakage of a chromosome, which can lead to four types of changes in chromosome structure **(Figure 15.14)**. A **deletion** occurs when a chromosomal fragment is lost. The affected chromosome is then missing certain genes. A broken fragment may become reattached as an extra segment to a sister or nonsister chromatid, producing a **duplication** of a portion of that chromosome. A chromosomal fragment may also reattach to the original chromosome but in the reverse orientation, producing an **inversion**. A fourth possible result of chromosomal breakage is for the fragment to join a nonhomologous chromosome, a rearrangement called a **translocation**.

Figure 15.14

```
    Alterations of chromosome structure.
```


(a) Deletion
A B C D E F G H
A **deletion** removes a chromosomal segment.
A B C E F G H
(b) Duplication
A B C D E F G H
A **duplication** repeats a segment.
A B C B C D E F G H
(c) Inversion
A B C D E F G H
An **inversion** reverses a segment within a chromosome.
A D C B E F G H
(d) Translocation
A B C D E F G H
M N O P Q R
A **translocation** moves a segment from one chromosome to a nonhomologous chromosome. In a reciprocal translocation, the most common type, nonhomologous



Red arrows indicate breakage points. Dark purple highlights the chromosomal parts affected by the rearrangements.

The parts are as follows.

Part a. Deletion. A deletion removes a chromosomal segment. A chromosome is drawn with the nucleotide sequence A B C D E F G H; red arrows indicate that the D is removed. The resulting chromosome reads A B C E F G H.

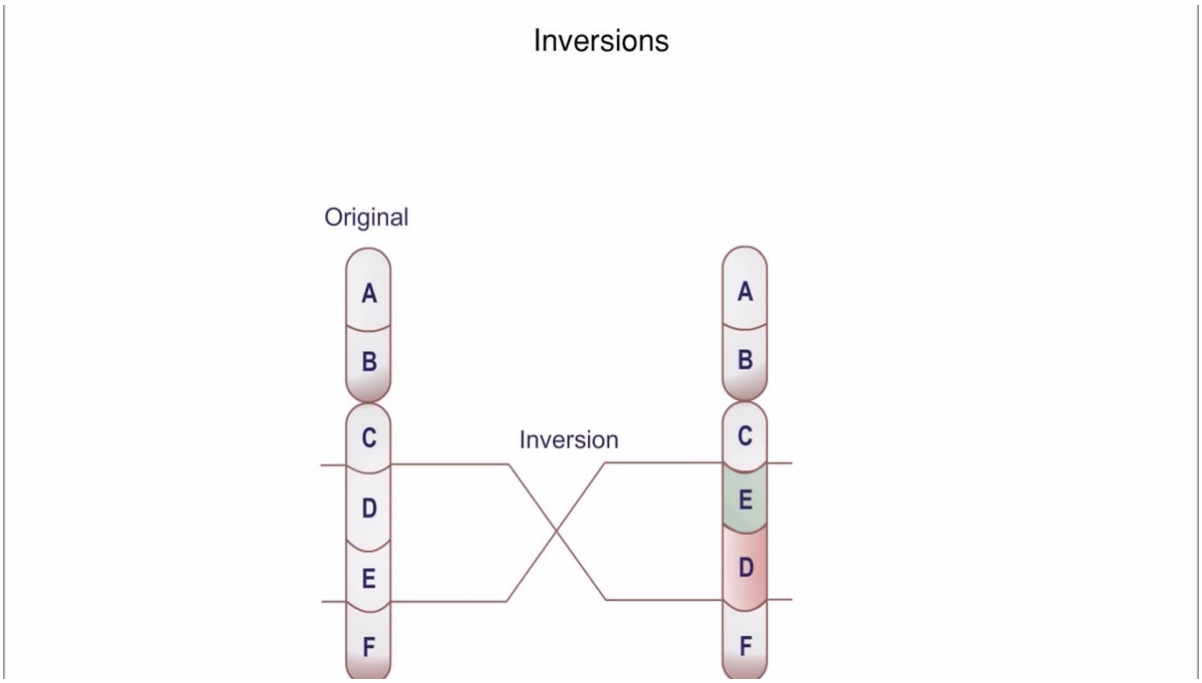
Part b. Duplication. A duplication repeats a segment. Chromosome drawn with nucleotide sequence A B C D E F G H; red arrows indicate a duplication with the resulting chromosome reading A B C B C D E F G H.

Part c. Inversion. An inversion reverses a segment within a chromosome. Chromosome with sequence A B C D E F G H; red arrows point to resulting chromosome: A D C B E F G H.

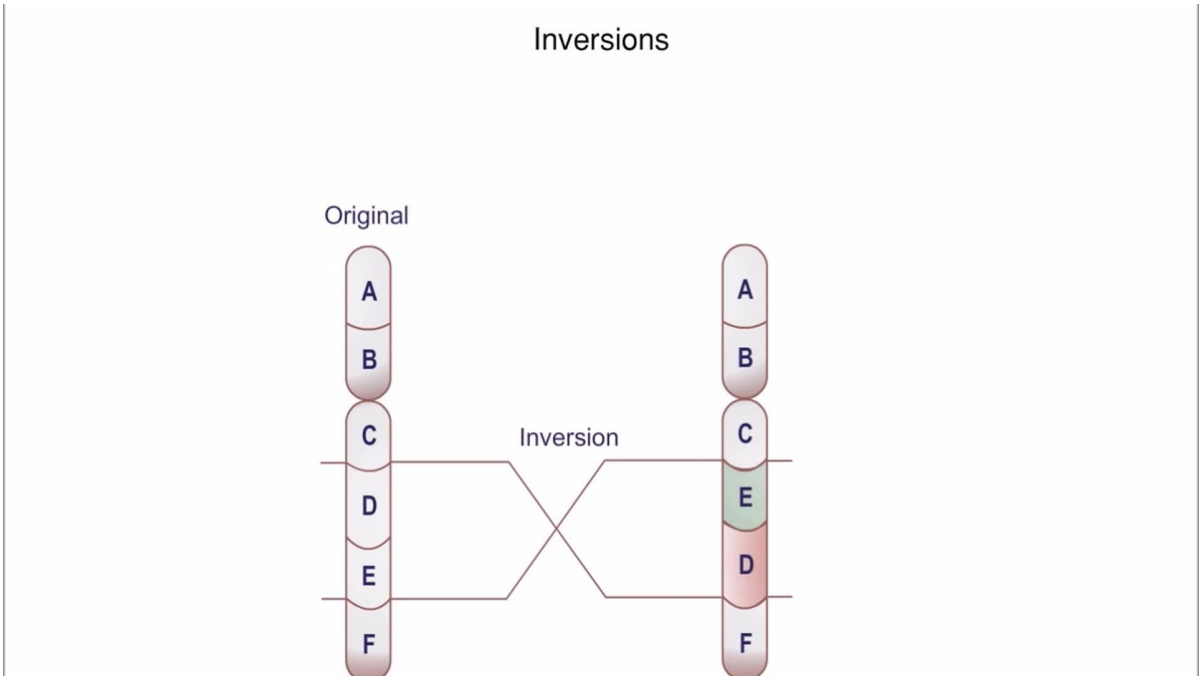
Part d. Translocation. A translocation moves a segment from one chromosome to a nonhomologous chromosome. In a reciprocal translocation, the most common type, nonhomologous chromosomes exchange fragments. Less often, nonreciprocal translocation occurs. Reciprocal translocation: Chromosomes with sequences A B C D E F G H and M N O P Q R; red arrows point to resulting chromosomes with sequences M N O C D E F H G and A B P Q R. Nonreciprocal translocation is not shown; in this case, a chromosome transfers a fragment but receives none in return.

Animation

Alterations of Chromosome Structure



Play



Play

Deletions and duplications are especially likely to occur during meiosis. In crossing over, nonsister chromatids sometimes exchange unequal-sized segments of DNA, so that one partner gives up more genes than it receives (see Figure 21.13). The products of such an unequal crossover are one chromosome with a deletion and one chromosome with a duplication.

A diploid embryo that is homozygous for a large deletion (both homologs have the deletion) or, if a male, has a single X chromosome with a large deletion is usually missing a number of essential genes. This condition is typically lethal.

Duplications and translocations also tend to be harmful. In reciprocal translocations, in which segments are exchanged between nonhomologous chromosomes, and in inversions, the balance of genes is not abnormal—all genes are present in their normal doses. Nevertheless, translocations and inversions can alter phenotype because a gene's expression can be influenced by its location among neighboring genes, which can have devastating effects.

Human Conditions Due to Chromosomal Alterations

Alterations of chromosome number and structure are associated with a number of human conditions, which vary in severity. As described earlier, nondisjunction in meiosis results in aneuploidy in gametes and any resulting zygotes. Although the frequency of aneuploid zygotes may be quite high in humans, most of these chromosomal alterations are so disastrous to development that the affected embryos are spontaneously aborted long before birth. However, some types of aneuploidy appear to upset the genetic balance less than others, so individuals with certain aneuploid conditions can survive to birth and beyond. These individuals have a set of traits—a *syndrome*—characteristic of the type of aneuploidy. Genetic disorders caused by aneuploidy can be diagnosed before birth by fetal testing (see Figure 14.19).

Down Syndrome (*Trisomy 21*)

One aneuploid condition, **Down syndrome**, affects approximately one out of every 830 children born in the United States **(Figure 15.15)**. Down syndrome is usually the result of an extra chromosome 21, so that each body cell has a total of 47 chromosomes. Because the cells are trisomic for chromosome 21, Down syndrome is often called *trisomy 21*. Down syndrome includes characteristic facial features, short stature, correctable heart defects, and developmental delays. Individuals with Down syndrome have an increased chance of developing leukemia and Alzheimer's disease but have a lower rate of high blood pressure, atherosclerosis (hardening of the arteries), stroke, and many types of solid tumors. People with Down syndrome have an average life span that is shorter than is typical; however, most live to middle age and beyond with proper medical care. Many live independently or at home with their families, are employed, and are valuable contributors to their communities. Almost all males and about half of females with Down syndrome are sexually underdeveloped and lack the ability to reproduce.

Figure 15.15

```
Down syndrome.
```

The karyotype shows trisomy 21, the most common cause of Down syndrome. The child exhibits the facial features characteristic of this syndrome.

The frequency of Down syndrome increases with the age of the female parent. While the syndrome occurs in just 0.1% (1 out of 900) births of children born to females at age 30, the risk climbs to 1% (1 out of 100) for females at age 40 and is even higher for older female parents. The correlation of Down syndrome with maternal age has not yet been explained. Most cases result from nondisjunction during meiosis I, and some research points to an age-dependent abnormality in meiosis. Trisomies of some other chromosomes (13 and 18) also increase in incidence with maternal age, although infants with other autosomal trisomies rarely survive for long. Medical experts recommend that prenatal screening for trisomies in the embryo be offered to all pregnant patients due to its low risk and useful results. In 2008, a law was passed stipulating that medical practitioners give accurate, up-to-date information about any prenatal or postnatal diagnosis received by parents and that they connect parents with appropriate support services.

Aneuploidy of Sex Chromosomes

Aneuploid conditions involving sex chromosomes appear to upset the genetic balance less than those involving autosomes. This may be because the Y chromosome carries relatively few genes. Also, extra copies of the X chromosome simply become inactivated as Barr bodies.

An extra X chromosome in a male, producing XXY, occurs approximately once in every 650 live male births. People with this condition, called *Klinefelter syndrome*, have male sex organs, but their testes are small and produce little or no sperm. Signs and symptoms vary widely, but may include taller than average height, less muscle mass, and enlarged breast tissue. Affected individuals may have learning disabilities. Testosterone replacement therapy helps stimulate changes that typically occur at puberty in males. About one of every 1,000 males is born with an extra Y chromosome (XYY). These males undergo typical sexual development and do not exhibit any well-defined syndrome, but tend to be taller than average.

Females with trisomy X(XXX), which occurs once in approximately 1,000 live female births, are generally healthy and have no unusual physical features other than being slightly taller than average. Females with trisomy X are at risk for learning disabilities. Monosomy X, which is called *Turner syndrome*, occurs about once in every 2,500 female births and is the only known viable monosomy in humans. Although these X0 individuals are phenotypically female, they usually lack the ability to reproduce because their sex organs do not mature. When provided with estrogen replacement therapy, girls with Turner syndrome do develop secondary sex characteristics. Most have typical intelligence.

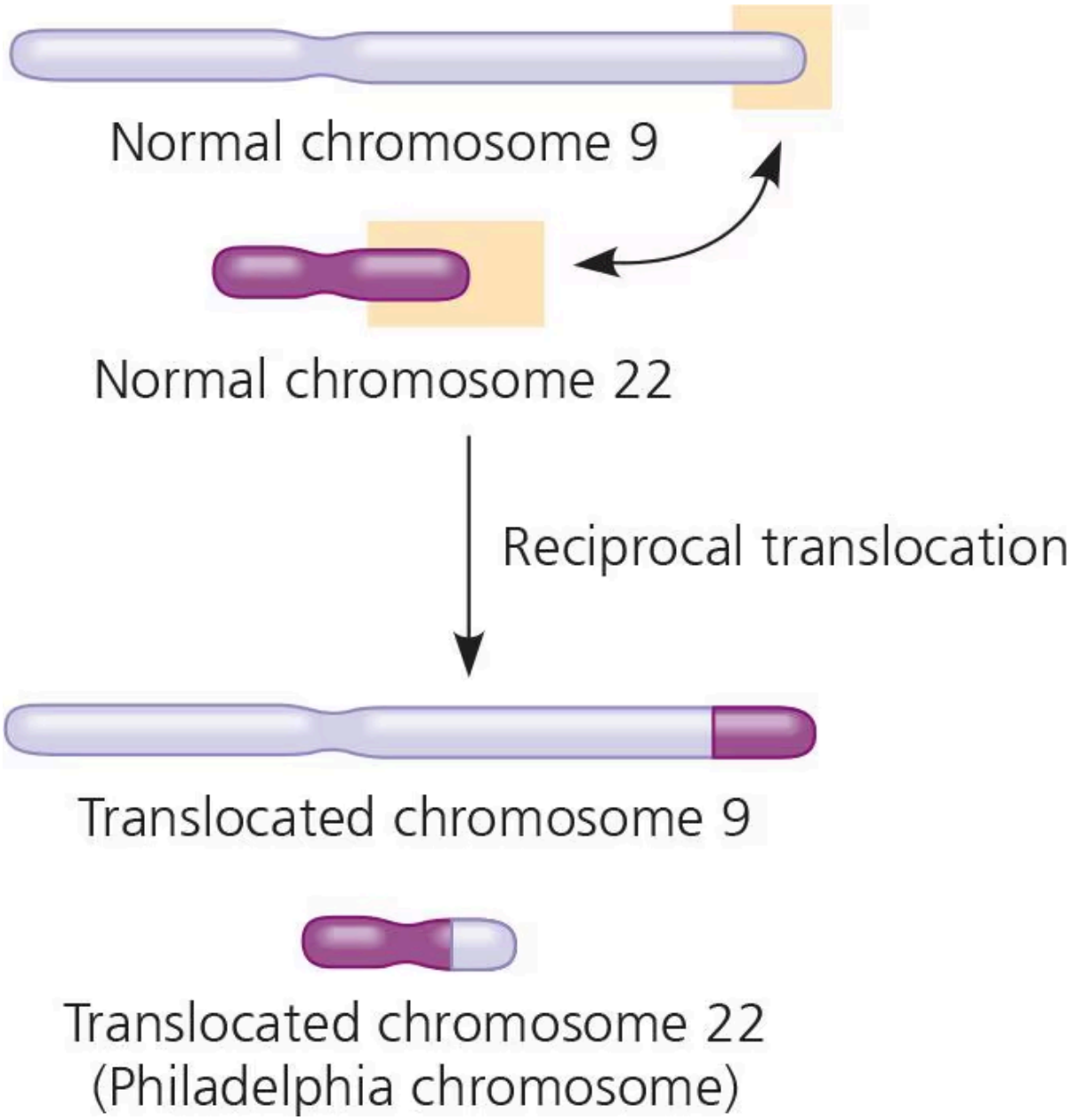
Disorders Caused by Structurally Altered Chromosomes

Many deletions in human chromosomes, even in a heterozygous state, cause severe problems. One such syndrome, known as *cri du chat* ("cry of the cat"), results from a specific deletion in chromosome 5. A child born with this deletion has a small head with unusual facial features, severe intellectual disabilities, and a cry that sounds like the mewing of a distressed cat. Such individuals usually die in infancy or early childhood.

Chromosomal translocations can also occur during mitosis; some have been implicated in certain cancers, including *chronic myelogenous leukemia*(*CML*). This disease occurs when a reciprocal translocation happens during mitosis of cells that are precursors of white blood cells. The exchange of a large portion of chromosome 22 with a small fragment from a tip of chromosome 9 produces a much shortened, easily recognized chromosome 22, called the *Philadelphia chromosome***(Figure 15.16)**. Such an exchange causes cancer by creating a new "fused" gene that leads to uncontrolled cell cycle progression. (The mechanism of gene activation will be discussed in Concept 18.5.)

Figure 15.16

```
Translocation associated with chronic myelogenous leukemia (CML).
```



The cancerous cells in nearly all CML patients contain an abnormally short chromosome 22, the so-called Philadelphia chromosome, and an abnormally long chromosome 9. These altered chromosomes result from the reciprocal translocation shown here, which presumably occurred in a single white blood cell precursor undergoing mitosis and was then passed along to all descendant cells.